Jesus Andres Ruiz

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Education

University of Texas at San Antonio — UTSA Honors College | ***GPA: 3.6***
Bachelor of Science in Mechanical Engineering — *Expected May 2028*

Skills

Software & Tools: SolidWorks, Fusion 360 (CAD + Simulation), FEA, MATLAB, Microsoft Office
Programming / Embedded: Arduino, Python, Git/GitHub, ROS (learning)
Manufacturing & Design: DFM, GD&T, Design Iteration, Rapid Prototyping, Failure/Root Cause Analysis
Fabrication & Tools: FDM & Continuous Carbon Fiber Printing, Wiring/Soldering, Mig/Tig Welding, CNC Machining
Materials & Testing: 6061-T6 Aluminum, Carbon Fiber Composites, Data Acquisition, Instrumentation, Schlieren Imaging

Relevant Experience

UTSA Intelligent Systems and Bionics Engineering Lab — San Antonio, TX
Undergraduate Research Assistant — *May 2026 – Present*

- Led fabrication of an exoskeleton hip module, working closely with graduate students to fabricate and assemble hardware across 3D printing and machining.
- Diagnosed and resolved 3D printer hardware/adhesion issues across 2 printer platforms (FDM and continuous carbon fiber), restoring consistent prototyping output.

UTSA Hypersonics Lab — San Antonio, TX
Undergraduate Research Volunteer — *January 2026 – May 2026*

- Fabricated precision diaphragms in the machine shop for use in a Mach 7.2 Ludwieg Tube wind tunnel, supporting high-speed flow experiments and ensuring reliable burst pressures for test runs.
- Assisted researchers during hypersonic wind tunnel testing, helping prepare instrumentation, monitor runs, and collect experimental data from the wind tunnel.
- Set up and aligned laser-based Schlieren imaging systems to visualize shock waves and boundary-layer structures in hypersonic flow experiments.

UTSA Formula Society of Automotive Engineers — San Antonio, TX
Mechanical Engineer - Chassis / Brakes — *August 2025 – Present*

- Machined a set of aluminum spacers on the mill and lathe to set caliper-to-rotor alignment on the 2025 competition vehicle, holding thickness tolerances required for even pad contact and consistent clearance.
- Modeled brake-system components in SolidWorks and produced dimensioned engineering drawings with GD&T callouts, applying design-for-manufacturing principles to keep parts producible on in-house shop equipment.
- Collaborated with a 5-person chassis and brakes sub-team to iterate designs under FSAE competition constraints, supporting fitment checks and assembly during vehicle build.

Project Experience

Exoskeleton Hip Module — San Antonio, TX
Undergraduate Research Assistant – Project Lead — *May 2026 – Present*

- Re-designed hip module components in Fusion 360 and validated redesigns using Fusion 360 static stress simulation following structural failure analysis of cracked cuffs, sliders, and struts on early prototypes.
- Manufactured parts via 3D printing (FDM, continuous carbon fiber), CNC machining, and outsourced cutting; corrected a 200-micron shoulder screw bore tolerance issue caused by anodizing buildup before final assembly.
- Iterated the hip abductor through 3 design/print revisions before finalizing it as a machined, anodized 6061-T6 aluminum part.

Bowden Cable Transmission – Ankle Actuation — San Antonio, TX
Undergraduate Research Assistant — *May 2026 – Present*

- Prototyped a Bowden cable transmission enabling remote ankle actuation, reducing distal leg mass and metabolic cost of walking.
- Designed, printed, and tested 2 TPU cable insert geometries (tapered vs. straight) to optimize routing and fit for cables.